Review 2 Commitment

1. Project Information

Project code	GFA26SE153
Project name	Laboratory Information Management Subsystem for Contract Testing Laboratory on Odoo Platform (Phân hệ Quản lý Thông tin Phòng kiểm nghiệm dịch vụ trên nền tảng Odoo)
Team	Nguyễn Nam Hoàng SE183190 (leader), Lê Sỹ Tuyền SE190797, Đinh Gia Huy SE193759, Phạm Anh Kiệt SE190051, Lê Nguyễn Hoàng Anh SE184416
Supervisor	Nguyễn Thế Hoàng (hoangnt2@fpt.edu.vn)
Review 2 date	24/10/2026
Reviewer 1	To be completed at Review 2
Reviewer 2	To be completed at Review 2
Version	1.1
Date	09/11/2026

2. Final Registration Form

Final version	2.0 (per Form BM03 – Record of Changes)
Approved at Review 2 (Yes/No)	Pending Review 2 confirmation
Attached file name	SWP490_GFA26SE153_R2_RecordOfChanges.docx

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-04	Portal cart, drafts and reorder	Customer	Main: A visitor fills a cart without signing in and submits it with their details. Rule: A package is ordered whole; reordering creates a draft at current prices and leaves the original unchanged. Exception: Submitting without an e-mail address is refused.	WF-01	Lê Sỹ Tuyền
F-05	Order claiming and orders on behalf	Sales	Main: A salesperson claims a submitted order, or creates one for a customer who called. Rule: A salesperson sees only their orders and the unclaimed queue; the Head of Sales sees all. Exception: An order already claimed cannot be claimed by another salesperson.	WF-01	Lê Sỹ Tuyền
F-06	Quotation and revisions	Sales	Main: Sales confirms the ways and the expected samples calculated from the sampling requirements, applies a percentage discount and sends the quotation. Rule: Each item copies its price, discount and tax rate, so a sent quotation never changes; a revision makes the old version read-only. Exception: A quotation without tests cannot be sent, nor one above the Head of Sales's thresholds before it is approved.	WF-01	Lê Sỹ Tuyền
F-09	Advance invoice and online payment	Customer	Main: Sending a quotation with an advance creates exactly one posted advance invoice; paying it accepts the quotation. Rule: Online payment accepts any amount up to the remaining balance; a bank transfer or cash payment counts once the accountant confirms it. Exception: A repeated PayOS notification records no second payment.	WF-01	Lê Sỹ Tuyền
F-10	Quotation approval and order confirmation	Sales, Head of Sales	Main: A quotation above the Head of Sales's thresholds is approved before it is sent; confirming the accepted order creates its test request with the calculated expected samples. Rule: Quotations within the thresholds and confirmed orders need no approval; the customer pays the balance before testing unless they may order on credit. Exception: The salesperson who requested approval cannot approve; a declined quotation goes back to Sales with the reason; a quotation awaiting approval cannot be edited.	WF-01	Lê Sỹ Tuyền
F-15	Test requests and lab codes	Sample delivery staff	Main: Staff create a test request with its samples, parameters, expected date and report delivery. Rule: Each sample gets a unique lab code, numbered again each year; a request staff create skips the paid step.	WF-02	Nguyễn Nam Hoàng

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
			Exception: Confirming without a sample, or a sample without a parameter, is refused; testers see only the lab code.		
F-16	Sample collection and handover record	Sample delivery staff, Customer	Main: Staff claim the collection task and plan its time; it appears on their schedule and to-do list, with a reminder. Rule: The customer confirms the actual samples and signs the handover record before labels are printed; signing stores the signature, time and device and e-mails the PDF. Exception: A signed handover record cannot be edited; a correction is a new record.	WF-02	Nguyễn Nam Hoàng
F-17	Reception, labels and department handover	Sample delivery staff, Head of department	Main: The condition is checked first; a sample in good condition is recorded as received with its time, amount and storage, and one testing task per sample test is queued. Rule: The label shows the lab code, request code, due date and QR code, never the customer. Exception: A label scanned by a person outside the department is refused.	WF-02	Nguyễn Nam Hoàng
F-18	Insufficient and rejected samples	Sample delivery staff, Head of department	Main: An insufficient sample goes on hold before receipt and its head of department is told. Rule: Asking for more creates a collection task recording who asked and when; more sample returns it to received. Exception: An unusable sample is rejected by the system with its condition note and gets no testing task.	WF-02	Nguyễn Nam Hoàng
F-23	Task assignment and claiming	Head of department, Tester	Main: Assigning a sample assigns all its testing tasks in the department to one person, who is notified. Rule: An assigned task leaves the queue and appears in the assignee's to-do list. Exception: A claimed task cannot be claimed again; another department's task cannot be claimed; reassigning without a reason is refused.	WF-03	Phạm Anh Kiệt
F-24	Booking proposal and head approval	Head of department, Tester	Main: The system proposes a booking of the agreed way on the fit machine that finishes soonest before the due date; when it is feasible, the head approves it, the tester is notified, and the tester, required locations and machine are reserved together. Rule: A booking takes the run time rounded up to 15-minute slots; urgent tests first, then first in first out. Exception: Bookings of one machine never overlap; with no feasible booking the head reviews other dates, planned OverTime or outsourcing.	WF-03	Phạm Anh Kiệt
F-25	Schedules and reminders	Tester, Sample delivery staff	Main: A booking appears on the person's calendar, the machine's schedule and the to-do list. Rule: A reminder arrives 15 minutes before it starts, once on each channel. Exception: An event the administrator turned off notifies nobody; a mandatory event cannot be turned off.	WF-03	Phạm Anh Kiệt
F-30	Testing method steps	Administrator, Tester	Main: The tester sees the method's steps with number, duration, what to do and note, in number order. Rule: Two steps of one method cannot share a number. Exception: A tester cannot change a step.	WF-04	Đinh Gia Huy
F-32	Regulations and limits	Administrator, Tester	Main: When a regulation comparison is requested, the result shows the related limit copied with its unit and is assessed. Rule: Later changes to the regulation leave existing results unchanged; a regulation holds one limit per parameter. Exception: A result with a requested comparison cannot be sent without its assessment.	WF-04	Đinh Gia Huy
F-34	Result entry and approval	Tester, Head of department	Main: The tester enters, signs and sends a result, which closes the testing task; the head approves it. Rule: A rejected result is kept, and a retest adds a new result with a new booking; only the latest is sent for approval. Exception: A tester cannot approve their own result, but the head may approve an entry they made; a rejection opens a new testing task and tells the tester the reason.	WF-04	Đinh Gia Huy
F-35	Unusual result warning	Tester	Main: A numeric result far from approved results of the same parameter, way of testing	WF-04	Đinh Gia Huy

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
			and sample type warns before sending. Rule: A result near the regulation limit warns; with too few approved results only the limit check runs. Exception: Sending anyway keeps the warning, which the approving head sees; the warning never names a customer.		
F-36	Sample test cancellation	Head of department	Main: The head cancels a waiting, assigned or testing sample test with a reason; its task and bookings are cancelled. Rule: A sample whose remaining tests are approved completes; a cancelled test is left out of the report. Exception: Cancelling without a reason, or a test with an approved result, is refused.	WF-04	Đinh Gia Huy
F-38	Outsourcing suggestions, urgent outsourcing and dispatches	Head of department, Sample delivery staff	Main: Accepting an outsourcing suggestion, or outsourcing a test urgently with a reason, puts its sample tests in a dispatch; when staff carry it, the system records the date sent, who took it and the expected date. Rule: Only a subcontracted way with the test's parameter and method can be chosen; the customer is neither told nor asked. Exception: A dispatch whose tests have results cannot be cancelled; when a delivery fails, the dispatch, still draft, is cancelled, its tests stay outsourced and the same sample goes back to sample reception.	WF-05	Lê Nguyễn Hoàng Anh
F-40	Subcontractor results	Subcontractor	Main: The subcontractor receives the samples and enters the results of the tests sent to it, and the head is notified. Rule: The result shows that user as its author and goes straight to the head for approval like any other. Exception: A subcontractor user sees only its own dispatches and never a sample's customer.	WF-05	Lê Nguyễn Hoàng Anh
F-43	Report signing and digital signature	Lab head	Main: When every test of the request is approved or cancelled, the system creates the read-only draft report, validates its data, reserves a number and prepares the PDF in the request's language; the lab head reviews it, the PDF is digitally signed and stored, and issue makes the number official. Rule: The report prints the approving heads and the lab head, not the testers, and leaves out cancelled tests. Exception: With Viettel Sign down nothing is marked signed and signing is retried; rejecting without a reason or a named result is refused.	WF-06	Phạm Anh Kiệt
F-45	Report delivery and download	Sample delivery staff, Customer	Main: Staff record the delivery of an issued report; an e-mail delivery sends the PDF. Rule: An issued report waits in the delivery list until its delivery is recorded. Exception: An unissued report cannot be delivered; a customer downloads only their own reports.	WF-06	Phạm Anh Kiệt
F-47	Notifications and preferences	Customer	Main: An issued report notifies the customer on the channels they chose. Rule: A new customer is notified by e-mail without configuring anything; a changed preference applies to the next notification. Exception: A mandatory notification cannot be turned off.	WF-06	Phạm Anh Kiệt
F-53	Laboratory folders and sync	Tester	Main: Saving a file in a folder bound in Medilab Sync adds a version with the tester, the computer, the size and the hash. Rule: Only files matching the folder's sync rules are uploaded; content the file already has is not stored twice. Exception: A person who is not a member of the folder, or cannot see its sample, gets nothing from it.	WF-04	Đinh Gia Huy
F-54	File versions, conflicts and restore	Tester	Main: Restoring version 3 of a file with 5 versions adds version 6 with the content of version 3. Rule: A file uploaded from an older version is kept as a conflict version and the current version stays. Exception: No one can change or delete a version, its hash or its size.	WF-04	Đinh Gia Huy
F-57	Sample folders and signing file versions	Head of department	Main: Creating a sample's folders adds its raw-data and result folders. Rule: A signature belongs to one version and its hash; a newer version shows as not signed. Exception: Creating the folders twice adds no second	WF-04	Đinh Gia Huy

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
			folder; rejecting a version without a reason is refused.		

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-01	Performance	95th percentile response time of common operations	≤ 3 s with 30 concurrent users	Load test with k6 on staging	Lê Sỹ Tuyền
NFR-02	Availability	Time until the system answers again after one site is lost	≤ 60 s	Failover drill on staging, checking the health address every 5 s	Lê Sỹ Tuyền
NFR-03	Recoverability	Data lost, and time to restore from backups	≤ 5 min; ≤ 1 h	Restore drill into a new database cluster	Lê Sỹ Tuyền
NFR-04	Security	Protected actions that succeed without permission; secrets found in the repository	0; 0	Permission tests per role in CI; GitGuardian scan on every pull request	Phạm Anh Kiệt
NFR-05	Auditability	Signatures, approvals or deletions missing from the audit report	0	Audit tests in CI	Lê Nguyễn Hoàng Anh
NFR-06	Maintainability	Pull requests merged with a failing check: SonarQube Cloud quality gate, branch and title conventions, linting, type checking, unit and browser tests	0	GitHub requires every check to pass before a merge to main	Lê Sỹ Tuyền

5. Out of Scope

Item	Reason
Invoices through an e-invoice provider	Invoices are Odoo invoices; a certified provider contract is outside the project.
Several laboratories in one installation	Each installation serves one laboratory company.
Customer price lists	Special prices are given as quotation discounts.
Odoo Enterprise apps	Every module installs on free Odoo Community.

6. Workflow Commitments

6.1 Workflow WF-01

Workflow name	WF-01 – Order to confirmed test request
Owner (one member)	Lê Sỹ Tuyền
Supporting members	Phạm Anh Kiệt (PayOS online payment F-09, permissions)
Actors	Customer, Sales, Head of Sales, Accountant
Trigger	A customer submits a cart on the portal, or sales creates an order for a customer
Business outcome	Order confirmed; test request paid with its expected samples
Main entity state path	Quotation draft → sent → accepted, approved by the Head of Sales before it is sent when above the thresholds; order submitted → claimed → confirmed; test request draft → confirmed → paid
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-01

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-04	Portal cart, drafts and reorder	Main: A visitor fills a cart without signing in and submits it with their details. Rule: A package is ordered whole; reordering creates a draft at current prices and leaves the original unchanged. Exception: Submitting without an e-mail address is refused.	TC-F04-01..03
F-05	Order claiming and orders on behalf	Main: A salesperson claims a submitted order, or creates one for a customer who called. Rule: A salesperson sees only their orders and the unclaimed queue; the Head of Sales sees all. Exception: An order already claimed cannot be claimed by another salesperson.	TC-F05-01..03
F-06	Quotation and revisions	Main: Sales confirms the ways and the expected samples calculated from the sampling requirements, applies a percentage discount and sends the quotation. Rule: Each item copies its price, discount and tax rate, so a sent quotation never changes; a revision makes the old version read-only. Exception: A quotation without tests cannot be sent, nor one above the Head of Sales's thresholds before it is approved.	TC-F06-01..03
F-09	Advance invoice and online payment	Main: Sending a quotation with an advance creates exactly one posted advance invoice; paying it accepts the quotation. Rule: Online payment accepts any amount up to the remaining balance; a bank transfer or cash payment counts once the accountant confirms it. Exception: A repeated PayOS notification records no second payment.	TC-F09-01..03
F-10	Quotation approval and order confirmation	Main: A quotation above the Head of Sales's thresholds is approved before it is sent; confirming the accepted order creates its test request with the calculated expected samples. Rule: Quotations within the thresholds and confirmed orders need no approval; the customer pays the balance before testing unless they may order on credit. Exception: The salesperson who requested approval cannot approve; a declined quotation goes back to Sales with the reason; a quotation awaiting approval cannot be edited.	TC-F10-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	Main	Demo customer; 2 parameters in the catalog; PayOS sandbox	Submit cart → sales claims the order, confirms the ways and 1 expected sample, applies a discount above the threshold and requests approval → Head of Sales signs → sales sends with full advance → customer pays → sales confirms the order	Order confirmed; test request paid with 1 expected sample	Lê Sỹ Tuyền
DS-01b	Exception · WF-01-E1	Sent quotation	Customer declines with a reason → sales revises and sends the new version → customer pays	Old version read-only; order goes on with the revision	Lê Sỹ Tuyền
DS-01c	Exception · WF-01-E2	Quotation past its payment deadline	Sales extends the deadline → customer pays	Expired flag cleared; quotation accepted	Lê Sỹ Tuyền
DS-01d	Exception · WF-01-E3	Quotation with a discount above the threshold waiting for approval	Head of Sales rejects with a reason → sales adjusts the discount and requests again → Head of Sales signs	Sales told the reason; quotation approved, then sent	Lê Sỹ Tuyền

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01e	Alternative · WF-01-A4	Quotation sent with full advance	Customer pays by bank transfer → sales records the payment → accountant confirms it → quotation accepted → sales confirms the order	Payment counted once confirmed; order confirmed and test request paid	Lê Sỹ Tuyền

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	15	15	Lê Sỹ Tuyền, Phạm Anh Kiệt
End-to-end workflow test (≥ 1)	1	1	Lê Sỹ Tuyền
State transition & NFR tests (18 state + 4 NFR: NFR-01..03, 06)	22	19	Lê Sỹ Tuyền
Total for this workflow	38	35	Lê Sỹ Tuyền

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	4 / 4 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

6.2 Workflow WF-02

Workflow name	WF-02 – Sample reception and handover
Owner (one member)	Nguyễn Nam Hoàng
Supporting members	Đinh Gia Huy (test request lab codes F-15, sample states)
Actors	Customer, Sample delivery staff, Head of department
Trigger	Samples arrive at the laboratory, or sample delivery staff collect them from the customer
Business outcome	Samples coded, labelled and received; tasks queued in each department
Main entity state path	Sample expected → received, or → on hold → received, or → rejected, decided by its condition before receipt; test request paid → sample received
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-02

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-15	Test requests and lab codes	Main: Staff create a test request with its samples, parameters, expected date and report delivery. Rule: Each sample gets a unique lab code, numbered again each year; a request staff create skips the paid step. Exception: Confirming without a sample, or a sample without a parameter, is refused; testers see only the lab code.	TC-F15-01..03
F-16	Sample collection and handover record	Main: Staff claim the collection task and plan its time; it appears on their schedule and to-do list, with a reminder. Rule: The customer confirms the actual samples and signs the handover record before labels are printed; signing stores the signature, time and device and e-mails the PDF. Exception: A signed handover record cannot be edited; a correction is a new record.	TC-F16-01..03
F-17	Reception, labels and department handover	Main: The condition is checked first; a sample in good condition is recorded as received with its time, amount and storage, and one testing task per sample test is queued. Rule: The label shows the lab code, request code, due date and QR code, never the customer. Exception: A label scanned by a person outside the department is refused.	TC-F17-01..03
F-18	Insufficient and rejected samples	Main: An insufficient sample goes on hold before receipt and its head of department is told. Rule: Asking for more creates a collection task recording who asked and when; more sample returns it to received. Exception: An unusable sample is rejected by the system with its condition note and gets no testing task.	TC-F18-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	Main	Paid request with 1 expected sample	Customer hands over the sample → staff record receipt in good condition → customer confirms and signs the handover record → signed PDF e-mailed → print label → department scans label	Sample received; task queued; signed PDF e-mailed	Nguyễn Nam Hoàng
DS-02b	Exception · WF-02-E1	Expected sample arrives insufficient	Record it insufficient → head asks for more → staff collect through the collection task → record receipt	Sample on hold, then received; tasks queued	Nguyễn Nam Hoàng
DS-02c	Exception · WF-02 unusable sample	Request with two samples, one unusable	Record the unusable one as damaged with a condition note → system rejects it	That sample rejected with its note; the other goes on to testing	Nguyễn Nam Hoàng
DS-02d	Exception · WF-02-E2	Signed handover record with a wrong quantity	Staff make a corrected record → customer signs it	New record signed; the old one kept unchanged	Nguyễn Nam Hoàng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	12	12	Nguyễn Nam Hoàng, Đinh Gia Huy
End-to-end workflow test (≥ 1)	1	1	Nguyễn Nam Hoàng
State transition tests (≥ 1 per transition)	11	11	Nguyễn Nam Hoàng
Total for this workflow	24	24	Nguyễn Nam Hoàng

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	3 / 3 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

6.3 Workflow WF-03

Workflow name	WF-03 – Testing assignment and scheduling
Owner (one member)	Phạm Anh Kiệt
Supporting members	Đinh Gia Huy (machine scheduler D4), Lê Sỹ Tuyền (tasks and reminders)
Actors	Head of department, Tester
Trigger	A testing task enters the department queue
Business outcome	Test assigned and booked in its agreed way on the fit machine that finishes it soonest before its due date, after the head approves the booking
Main entity state path	Sample test waiting → assigned → testing; booking proposed → scheduled → in progress → done
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-03

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-23	Task assignment and claiming	Main: Assigning a sample assigns all its testing tasks in the department to one person, who is notified. Rule: An assigned task leaves the queue and appears in the assignee's to-do list. Exception: A claimed task cannot be claimed again; another department's task cannot be claimed; reassigning without a reason is refused.	TC-F23-01..03
F-24	Booking proposal and head approval	Main: The system proposes a booking of the agreed way on the fit machine that finishes soonest before the due date; when it is feasible, the head approves it, the tester is notified, and the tester, required locations and machine are reserved together. Rule: A booking takes the run time rounded up to 15-minute slots; urgent tests first, then first in first out. Exception: Bookings of one machine never overlap; with no feasible booking the head reviews other dates, planned OverTime or outsourcing.	TC-F24-01..03
F-25	Schedules and reminders	Main: A booking appears on the person's calendar, the machine's schedule and the to-do list. Rule: A reminder	TC-F25-01..03

Feature ID	Feature	Acceptance criteria	Test case IDs
		arrives 15 minutes before it starts, once on each channel. Exception: An event the administrator turned off notifies nobody; a mandatory event cannot be turned off.	

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	Main	Received sample; two fit machines with different free slots	Head assigns the sample → tester is notified and views the proposed booking → head approves it → tester is notified → tester, locations and machine reserved	Booking on the machine that finishes first, before the due date, on the tester's schedule	Phạm Anh Kiệt
DS-03b	Exception · WF-03-E1	Task assigned to a tester who is absent	Head moves it to another tester with a reason → new tester notified → head approves the proposed booking	Reason kept on the task; test booked in time	Phạm Anh Kiệt
DS-03c	Exception · WF-03-E2	The only machine for the test is overdue for calibration	System leaves out the overdue machine and finds no feasible slot → suggests outsourcing → head accepts it	No booking; test outsourced	Phạm Anh Kiệt

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	9	9	Phạm Anh Kiệt, Đinh Gia Huy
End-to-end workflow test (≥ 1)	1	1	Phạm Anh Kiệt
State transition & NFR tests (10 state + 1 NFR: NFR-04)	11	11	Phạm Anh Kiệt
Total for this workflow	21	21	Phạm Anh Kiệt

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	2 / 2 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

6.4 Workflow WF-04

Workflow name	WF-04 – Result entry and approval
Owner (one member)	Đinh Gia Huy
Supporting members	Phạm Anh Kiệt (laboratory folders, file versions and sync F-53, F-54, F-57)
Actors	Tester, Head of department
Trigger	A booked test starts; its machine is marked busy until the testing task closes
Business outcome	Result approved against its limit; the result file kept as a signed version; sample completed
Main entity state path	Sample test testing → result entered → approved (→ testing when rejected); sample testing → awaiting approval → completed; result file version current or conflict, then signed; machine available → busy → available
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-04, §SF-11

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-30	Testing method steps	Main: The tester sees the method's steps with number, duration, what to do and note, in number order. Rule: Two steps of one method cannot share a number. Exception: A tester cannot change a step.	TC-F30-01..03
F-32	Regulations and limits	Main: When a regulation comparison is requested, the result shows the related limit copied with its unit and is assessed. Rule: Later changes to the regulation leave existing results unchanged; a regulation holds one limit per parameter. Exception: A result with a requested comparison cannot be sent without its assessment.	TC-F32-01..03

Feature ID	Feature	Acceptance criteria	Test case IDs
F-34	Result entry and approval	Main: The tester enters, signs and sends a result, which closes the testing task; the head approves it. Rule: A rejected result is kept, and a retest adds a new result with a new booking; only the latest is sent for approval. Exception: A tester cannot approve their own result, but the head may approve an entry they made; a rejection opens a new testing task and tells the tester the reason.	TC-F34-01..03
F-35	Unusual result warning	Main: A numeric result far from approved results of the same parameter, way of testing and sample type warns before sending. Rule: A result near the regulation limit warns; with too few approved results only the limit check runs. Exception: Sending anyway keeps the warning, which the approving head sees; the warning never names a customer.	TC-F35-01..03
F-36	Sample test cancellation	Main: The head cancels a waiting, assigned or testing sample test with a reason; its task and bookings are cancelled. Rule: A sample whose remaining tests are approved completes; a cancelled test is left out of the report. Exception: Cancelling without a reason, or a test with an approved result, is refused.	TC-F36-01..03
F-53	Laboratory folders and sync	Main: Saving a file in a folder bound in Medilab Sync adds a version with the tester, the computer, the size and the hash. Rule: Only files matching the folder's sync rules are uploaded; content the file already has is not stored twice. Exception: A person who is not a member of the folder, or cannot see its sample, gets nothing from it.	TC-F53-01..03
F-54	File versions, conflicts and restore	Main: Restoring version 3 of a file with 5 versions adds version 6 with the content of version 3. Rule: A file uploaded from an older version is kept as a conflict version and the current version stays. Exception: No one can change or delete a version, its hash or its size.	TC-F54-01..03
F-57	Sample folders and signing file versions	Main: Creating a sample's folders adds its raw-data and result folders. Rule: A signature belongs to one version and its hash; a newer version shows as not signed. Exception: Creating the folders twice adds no second folder; rejecting a version without a reason is refused.	TC-F57-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04a	Main	Booked test on a sample with a regulation; the sample's result folder bound on the tester's computer	Booking starts and the machine is marked busy → tester saves the instrument's result file → Medilab Sync uploads it → head signs the file version → tester enters the result and its assessment → sends and signs → head approves the result	Result keeps the copied limit; file version signed; sample completed	Đinh Gia Huy
DS-04b	Exception · WF-04-E1	Result sent for approval	Head rejects with a reason → tester retests with a new booking → head approves the new result	Rejected result kept on record; new result approved	Đinh Gia Huy
DS-04c	Exception · WF-04-E2	Result far from past approved results	Tester sees the warning → sends anyway → head approves	Warning kept on the result and shown to the head	Đinh Gia Huy
DS-04d	Exception · WF-04-E3	Test that cannot be done on the sample	Head cancels it with a reason	Sample completes with its other tests; cancelled test left out of the report	Đinh Gia Huy
DS-04e	Exception · WF-04-E4	Same result file changed on two computers from one version	Second tester's upload becomes a conflict → they take the server's version → save their changes again	Both versions kept; local file backed up; changes saved again	Đinh Gia Huy
DS-04f	Exception · WF-04-E5	Result file version waiting for signature	Head rejects it with a reason → tester saves a corrected version → head signs	Corrected version signed	Đinh Gia Huy
DS-04g	Exception · WF-04-E6	Online editing service down	Tester saves the result file in its own program	Medilab Sync uploads it; history and restore keep working	Đinh Gia Huy

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	24	24	Đinh Gia Huy, Phạm Anh Kiệt
End-to-end workflow test (≥ 1)	1	1	Đinh Gia Huy
State transition tests (≥ 1 per transition)	16	16	Đinh Gia Huy, Phạm Anh Kiệt
Total for this workflow	41	41	Đinh Gia Huy

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	6 / 6 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

6.5 Workflow WF-05

Workflow name	WF-05 – Outsourced testing
Owner (one member)	Lê Nguyễn Hoàng Anh
Supporting members	Đinh Gia Huy (outsourcing suggestions from scheduler)
Actors	Head of department, Sample delivery staff, Subcontractor
Trigger	No machine can finish a test before its due date, or the head outsources a test urgently
Business outcome	Subcontractor's result approved
Main entity state path	Sample test waiting or assigned → outsourced → result entered → approved (→ outsourced when rejected); dispatch draft → sent → received, or draft → cancelled before any result
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-05

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-38	Outsourcing suggestions, urgent outsourcing and dispatches	Main: Accepting an outsourcing suggestion, or outsourcing a test urgently with a reason, puts its sample tests in a dispatch; when staff carry it, the system records the date sent, who took it and the expected date. Rule: Only a subcontracted way with the test's parameter and method can be chosen; the customer is neither told nor asked. Exception: A dispatch whose tests have results cannot be cancelled; when a delivery fails, the dispatch, still draft, is cancelled, its tests stay outsourced and the same sample goes back to sample reception.	TC-F38-01..03
F-40	Subcontractor results	Main: The subcontractor receives the samples and enters the results of the tests sent to it, and the head is notified. Rule: The result shows that user as its author and goes straight to the head for approval like any other. Exception: A subcontractor user sees only its own dispatches and never a sample's customer.	TC-F40-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05a	Main	Only machine for the test is under repair; accredited subcontractor seeded	System suggests outsourcing → head accepts it and puts the sample tests in a dispatch → staff carry the dispatch and the system records it → subcontractor receives the samples and enters the result → head approves	Dispatch received; sample test approved	Lê Nguyễn Hoàng Anh
DS-05b	Exception · WF-05-E1	Subcontractor's result waiting for approval	Head rejects it → subcontractor enters a new result → head approves	Test back to outsourced, then approved	Lê Nguyễn Hoàng Anh
DS-05c	Exception · WF-05-E2	Draft dispatch that cannot go out	Head cancels it → puts its tests in another dispatch	Tests stay outsourced and go with the new dispatch	Lê Nguyễn Hoàng Anh
DS-05d	Alternative · WF-05-A3	Machine breaks; test assigned with no outsourcing suggestion	Head requests outsourcing with a reason → puts the test in a dispatch → staff carry it to the subcontractor	Reason shown on the test; dispatch on its way to the subcontractor	Lê Nguyễn Hoàng Anh

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	6	6	Lê Nguyễn Hoàng Anh
End-to-end workflow test (≥ 1)	1	1	Lê Nguyễn Hoàng Anh
State transition & NFR tests (11 state + 1 NFR: NFR-05)	12	12	Lê Nguyễn Hoàng Anh
Total for this workflow	19	19	Lê Nguyễn Hoàng Anh

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	3 / 3 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

6.6 Workflow WF-06

Workflow name	WF-06 – Report signing, release and delivery
Owner (one member)	Phạm Anh Kiệt
Supporting members	Đinh Gia Huy (test report generation and signing chain)
Actors	Lab head, Head of department, Sample delivery staff, Customer
Trigger	Every sample test of a request is approved or cancelled
Business outcome	Signed report issued, delivered and downloadable on the portal
Main entity state path	Test report draft → signing → issued (→ replaced); test request completed → reported
Activity diagram (Report 4 section)	Report 4 / docs/functional/business-flows.md §WF-06

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-43	Report signing and digital signature	Main: When every test of the request is approved or cancelled, the system creates the read-only draft report, validates its data, reserves a number and prepares the PDF in the request's language; the lab head reviews it, the PDF is digitally signed and stored, and issue makes the number official. Rule: The report prints the approving heads and the lab head, not the testers, and leaves out cancelled tests. Exception: With Viettel Sign down nothing is marked signed and signing is retried; rejecting without a reason or a named result is refused.	TC-F43-01..03
F-45	Report delivery and download	Main: Staff record the delivery of an issued report; an e-mail delivery sends the PDF. Rule: An issued report waits in the delivery list until its delivery is recorded. Exception: An unissued report cannot be delivered; a customer downloads only their own reports.	TC-F45-01..03
F-47	Notifications and preferences	Main: An issued report notifies the customer on the channels they chose. Rule: A new customer is notified by e-mail without configuring anything; a changed preference applies to the next notification. Exception: A mandatory notification cannot be turned off.	TC-F47-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-06a	Main	Request with all results approved; Viettel Sign sandbox	System prepares the PDF → lab head reviews and approves it → PDF digitally signed and stored → report issued → staff records e-mail delivery → customer downloads	Report issued with number and signed PDF; request reported	Phạm Anh Kiệt
DS-06b	Exception · WF-06-E2	Draft report	Lab head rejects naming a result → head of department approves it again → lab head signs	Result back in the department queue, then the report issued	Phạm Anh Kiệt
DS-06c	Exception · WF-06-E1	Viettel Sign unreachable	Lab head approves → signing retried once Viettel Sign answers	Report not marked signed until the retry succeeds, then issued	Phạm Anh Kiệt

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (≥ 3 per feature)	9	9	Phạm Anh Kiệt, Đinh Gia Huy
End-to-end workflow test (≥ 1)	1	1	Phạm Anh Kiệt
State transition tests (≥ 1 per transition)	9	9	Phạm Anh Kiệt
Total for this workflow	19	19	Phạm Anh Kiệt

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	Yes
Main scenario passes	Yes (100%)
Exception scenarios passing	2 / 2 (100%; ≥ 80% threshold)
Test pass rate for this workflow	≥ 95% overall; 100% main-path tests of mandatory features
Open Critical / High defects	0

Reviewer note

Note	

7. Individual Commitments

7.1 Member 1

Full name	Nguyễn Nam Hoàng
Student ID	SE183190
Role in team	Team Leader & Developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-02 (Sample reception and handover)	sole owner
Mandatory features owned	F-15, F-16, F-17, F-18 (4 features)	sample reception, lab codes, QR labels, handover records
Modules / entities owned	sol_laboratory (SampleHandover, SampleReception, SampleCondition), sol_commerce (Customer Portal & Catalog)	reception & portal UI
Design decisions owned (D-IDs)	D1 (Platform and modules – Odoo 20 Community)	architecture & module boundaries
NFRs responsible	Supports NFR-01, NFR-04 on reception and portal	portal and reception verification

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	24 (TC-F15-01..03 to TC-F18-01..03, 1 E2E, 11 state transitions)	12/11/2026
Of which automated in CI	24 (100%)	12/11/2026
Demo scenarios presented	4 (DS-02a Main; DS-02b, DS-02c, DS-02d Exception)	Defense 1.1
Required fixes owned (section 9)	Coordinates team verification of RF-01 before Defense 1.1	12/11/2026
Documentation sections (Report / section)	Report 1 (Project Overview), Report 3 (Reception & Portal SRS), BM01/BM03/BM04 coordination	05/11/2026

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	Nguyễn Nam Hoàng –

7.2 Member 2

Full name	Lê Sỹ Tuyền
Student ID	SE190797
Role in team	DevOps & Backend Developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-01 (Order to confirmed test request)	sole owner
Mandatory features owned	F-04, F-05, F-06, F-09, F-10 (5 features)	portal cart, order claiming, quotation revisions, quotation approval
Modules / entities owned	sol_commerce (SaleOrder, QuotationVersion, OrderApproval, CancellationRequest); sol_laboratory (Schedule, TodoItem); Kubernetes & GitOps infra	orders, schedules & k3s/CNPG/Argo CD
Design decisions owned (D-IDs)	D5 (Deployment – Dual-site k3s, CloudNativePG, S3 backups, Argo CD)	high availability & CI/CD
NFRs responsible	NFR-01 (Performance), NFR-02 (Availability), NFR-03 (Recoverability), NFR-06 (Maintainability)	k6 load test, failover & restore drills, CI gates

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	38 (TC-F04-01..03 to TC-F10-01..03, 1 E2E, 18 state transitions, 4 NFR tests)	12/11/2026

Commitment	Committed (count / IDs)	Due
Of which automated in CI	35 (35 in GitHub Actions CI + 3 staging drills for NFR-01..03)	12/11/2026
Demo scenarios presented	5 (DS-01a Main; DS-01b, DS-01c, DS-01d Exception; DS-01e Alternative)	Defense 1.1
Required fixes owned (section 9)	RF-01 (price periods)	12/11/2026
Documentation sections (Report / section)	Report 3 (E-commerce SRS & NFRs), Report 4 (Architecture, Deployment & ERD diagrams)	05/11/2026

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	Lê Sỹ Tuyền –

7.3 Member 3

Full name	Đinh Gia Huy
Student ID	SE193759
Role in team	Lead Backend Developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-04 (Result entry and approval)	sole owner
Mandatory features owned	F-30, F-32, F-34, F-35, F-36, F-53, F-54, F-57 (8 features in WF-04; co-delivers F-53, F-54, F-57 with Phạm Anh Kiệt)	methods, regulation limits, results, retests, cancellations
Modules / entities owned	sol_laboratory (TestRequest, Sample, SampleTest, TestResult, Regulation, TestingMethod, MachineBooking)	core laboratory domain & scheduler
Design decisions owned (D-IDs)	D3 (Approving is signing), D4 (Machine scheduling in 15-minute slots)	signing chains & machine queue
NFRs responsible	Co-owns NFR-04 (Sample anonymisation R2) and NFR-05 (Result & signature auditability)	role & anonymisation tests

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	41 (TC-F30..F36, TC-F53, TC-F54, TC-F57, 1 E2E, 16 state transitions)	12/11/2026
Of which automated in CI	41 (100%)	12/11/2026
Demo scenarios presented	7 (DS-04a Main; DS-04b, DS-04c, DS-04d, DS-04e, DS-04f, DS-04g Exception)	Defense 1.1
Required fixes owned (section 9)	None	—
Documentation sections (Report / section)	Report 3 (Laboratory SRS), Report 4 (State machines, signing chain & scheduling algorithms)	05/11/2026

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	Đinh Gia Huy –

7.4 Member 4

Full name	Phạm Anh Kiệt
Student ID	SE190051
Role in team	Full-Stack & Systems Developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-03 (Testing assignment and scheduling), WF-06 (Report signing, release and delivery)	owns 2 main workflows
Mandatory features owned	F-23, F-24, F-25 (WF-03) and F-43, F-45, F-47 (WF-06) (6 workflow features + laboratory file sync infrastructure F-53, F-54, F-57)	assignment, scheduling, digital signing, delivery, notifications, Medilab Sync
Modules / entities owned	sol_laboratory (Role/Permission, LaboratoryFolder, LaboratoryFile, LaboratoryUpload, TestReport, Notification); sol_commerce (PayOS); medilab-sync (Rust/Tauri)	permissions, S3 sync, Viettel Sign, PayOS, notifications
Design decisions owned (D-IDs)	D2 (Role & permission matrix generating Odoo groups and record rules)	access control across UI & API
NFRs responsible	NFR-04 (Security – 0 unauthorized actions across roles; 0 secrets in repo)	permission test suite & GitGuardian

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	40 (21 in WF-03 + 19 in WF-06, including NFR-04 permission tests)	12/11/2026
Of which automated in CI	40 (100%)	12/11/2026
Demo scenarios presented	6 (DS-03a, DS-06a Main; DS-03b, DS-03c, DS-06b, DS-06c Exception)	Defense 1.1
Required fixes owned (section 9)	None	—
Documentation sections (Report / section)	Report 3 & Report 4 (Permissions spec, PayOS spec, Laboratory Files & Medilab Sync spec)	05/11/2026

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	Phạm Anh Kiệt –

7.5 Member 5

Full name	Lê Nguyễn Hoàng Anh
Student ID	SE184416
Role in team	Backend Developer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-05 (Outsourced testing)	sole owner
Mandatory features owned	F-38, F-40 (2 features)	outsourcing suggestions, urgent outsourcing, dispatches, subcontractor portal results
Modules / entities owned	sol_laboratory (Subcontractor, OutsourcingSuggestion, SubcontractDispatch, AuditLog); sol_inventory (Chemical, StockReceipt, StockIssue)	outsourcing, audit trail & chemical inventory
Design decisions owned (D-IDs)	Co-owns D3/SH-05 (Immutable audit log and subcontractor isolation)	audit trail & subcontractor access rules
NFRs responsible	NFR-05 (Auditability – 0 signatures, approvals or deletions missing from audit trail)	automated audit trail tests in CI

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	19 (TC-F38-01..03, TC-F40-01..03, 1 E2E, 11 state transitions, 1 NFR-05 test)	12/11/2026
Of which automated in CI	19 (100%)	12/11/2026
Demo scenarios presented	4 (DS-05a Main; DS-05b, DS-05c Exception; DS-05d Alternative)	Defense 1.1
Required fixes owned (section 9)	None	—
Documentation sections (Report / section)	Report 3 & Report 4 (Outsourced testing workflow WF-05, Audit trail SH-05, Inventory ERD)	05/11/2026

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ *Verified* ☐ *Partly verified* ☐ *Not verified*		

Member confirmation

Signature / date	Lê Nguyễn Hoàng Anh –

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	Lê Sỹ Tuyền	5	5 (1 / 4)	38	35
WF-02	Nguyễn Nam Hoàng	4	4 (1 / 3)	24	24
WF-03	Phạm Anh Kiệt	3	3 (1 / 2)	21	21
WF-04	Đinh Gia Huy	8	7 (1 / 6)	41	41
WF-05	Lê Nguyễn Hoàng Anh	2	4 (1 / 3)	19	19
WF-06	Phạm Anh Kiệt	3	3 (1 / 2)	19	19
Total	5 members (6 workflows)	25	26 (6 / 20)	162	159

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Nguyễn Nam Hoàng	WF-02	4	24	24	4 (1 / 3)	Pending Review 2
Lê Sỹ Tuyền	WF-01	5	38	35	5 (1 / 4)	Pending Review 2
Đinh Gia Huy	WF-04	8	41	41	7 (1 / 6)	Pending Review 2
Phạm Anh Kiệt	WF-03, WF-06	6	40	40	6 (2 / 4)	Pending Review 2
Lê Nguyễn Hoàng Anh	WF-05	2	19	19	4 (1 / 3)	Pending Review 2
Total	6 workflows	25	162	159	26 (6 / 20)	—

8.3 Committed Numbers

Number of main workflows	6 (WF-01 to WF-06)
Number of demo scenarios (main / exception)	26 total (6 main / 20 exception)
Committed functional test cases	162 (75 feature + 6 E2E workflow + 75 state transition + 6 NFR)
Of which automated in CI	159 (98.1%; 3 staging drills for NFR-01..03 run manually)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	100% run end-to-end	100% (all 6 workflows WF-01..WF-06 run end-to-end on staging)
Demo scenarios	100% main; ≥ 80% exception	100% main (6/6); ≥ 80% exception (≥ 16/20 scenarios)
Test case count	≥ committed number	≥ 162 test cases
Test pass rate	≥ 90% overall; 100% main-path tests of mandatory features	≥ 95% overall; 100% main-path tests of all 25 mandatory features
Automation	≥ 50% run in CI	98.1% (159 / 162 test cases run automatically in GitHub Actions CI)
Defects	No open Critical or High defects	0 open Critical or High defects
Individual	Each member demos the workflow they own and meets their delivery commitments	100% of members demo their owned workflow(s) and pass ownership verification

9. Required Fixes

ID	Item	Related workflow	Owner	Due
RF-01	Calculate and store prices by time period (Review 2 minutes): implement in sol_commerce the price periods of docs/functional/ecommerce_module/features/admin.md, recording each parameter and service package price with the period it applies (valid from, valid until, no overlapping periods), showing the price in force today in the catalog and cart, and pricing each quotation at the prices and tax rates in force on its date.	WF-01 (SF-08)	Lê Sỹ Tuyền	12/11/2026

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1			
Reviewer 2			
Supervisor	Nguyễn Thế Hoàng		
Team leader	Nguyễn Nam Hoàng		